Shape and Space 8 — Vectors · Column vectors · Vector geometry

Calculator allowed. Marks in brackets. ★ = harder question. Bold letters are vectors; $|\mathbf{a}|$ is the magnitude (length) of $\mathbf{a}$.

1 [5]

$\mathbf{p} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, $\mathbf{r} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$ and $2\mathbf{p} + \mathbf{r} = \mathbf{q} - \mathbf{s}$.

a Work out $\mathbf{s}$ as a column vector. **b** Calculate $|\mathbf{s}|$ to 1 d.p.

2 [8]

Points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 16 \\ 12 \end{pmatrix}$.

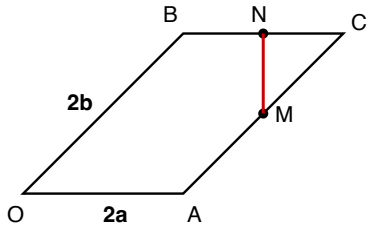
a Write down $\overrightarrow{AB}$. **b** Calculate $|\overrightarrow{AB}|$.

c Find the angle $\overrightarrow{AB}$ makes with the x-axis, to 1 d.p.

d X is the mid-point of AB. Write $\overrightarrow{OX}$ as a column vector.

3 [9]

OACB is a parallelogram with $\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OB} = 2\mathbf{b}$. M is the mid-point of AC and N is the mid-point of BC.

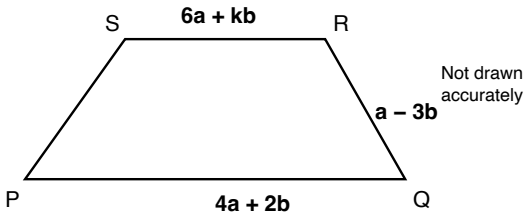


Express in terms of $\mathbf{a}$ and $\mathbf{b}$: **a** $\overrightarrow{OC}$ **b** $\overrightarrow{OM}$ **c** $\overrightarrow{ON}$ **d** $\overrightarrow{NM}$

e What do your answers tell you about NM and BA?

4 [3]

PQRS is a trapezium with PQ parallel to SR. $\overrightarrow{PQ} = 4\mathbf{a} + 2\mathbf{b}$, $\overrightarrow{QR} = \mathbf{a} - 3\mathbf{b}$ and $\overrightarrow{SR} = 6\mathbf{a} + k\mathbf{b}$, where k is a constant. Find k, showing your working.



Total: 25 marks

Check: add vectors component by component · $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ (end minus start) · magnitude by Pythagoras · a route around a shape: go along known vectors, backwards means a minus sign · parallel ⇔ one vector is a multiple of the other; same multiple through a shared point ⇔ collinear.